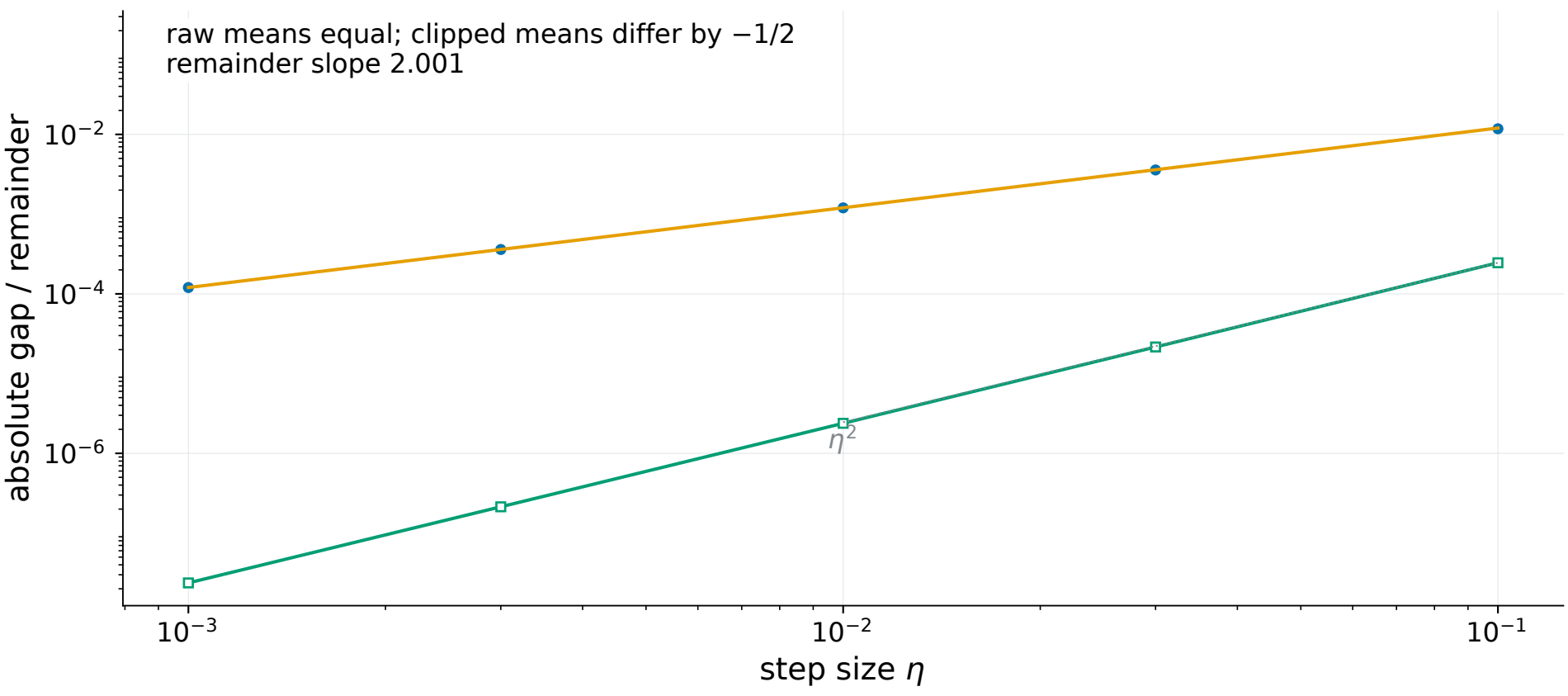
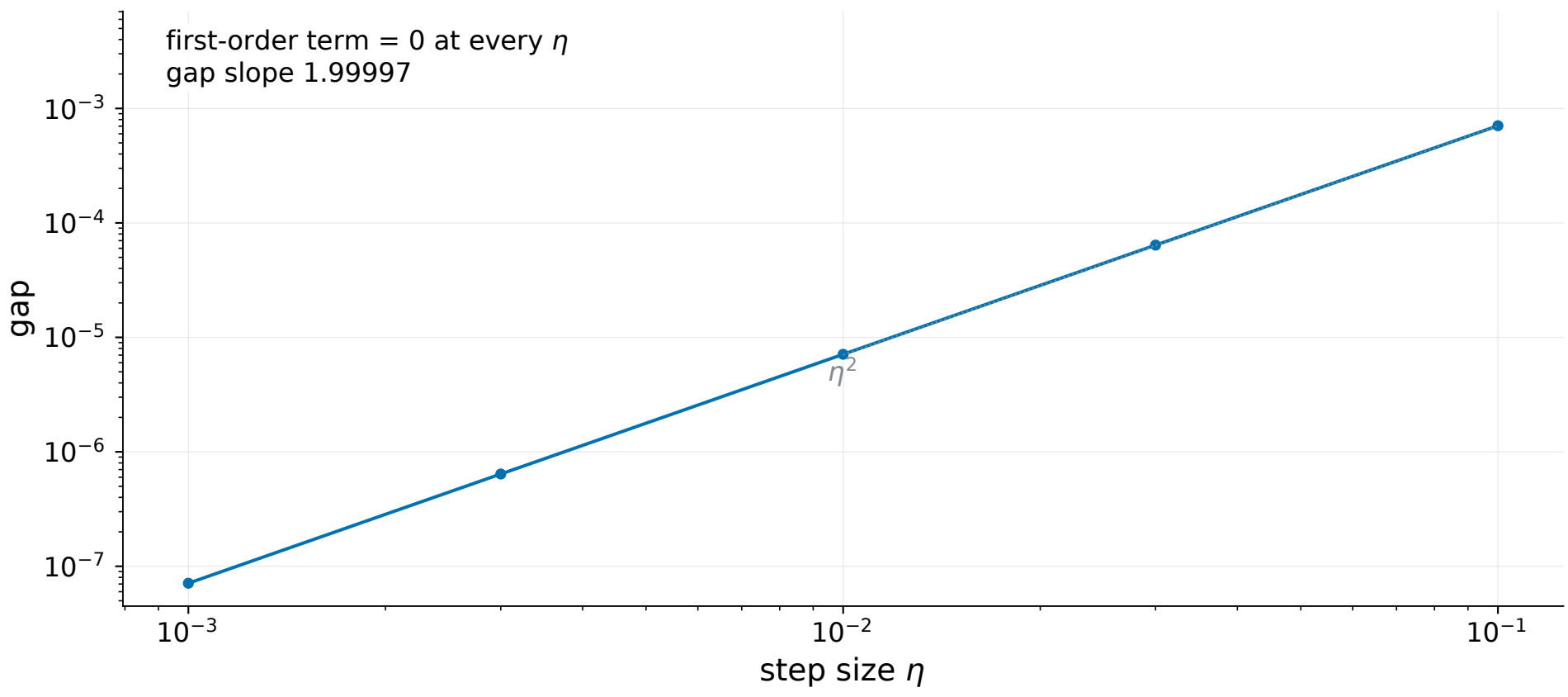
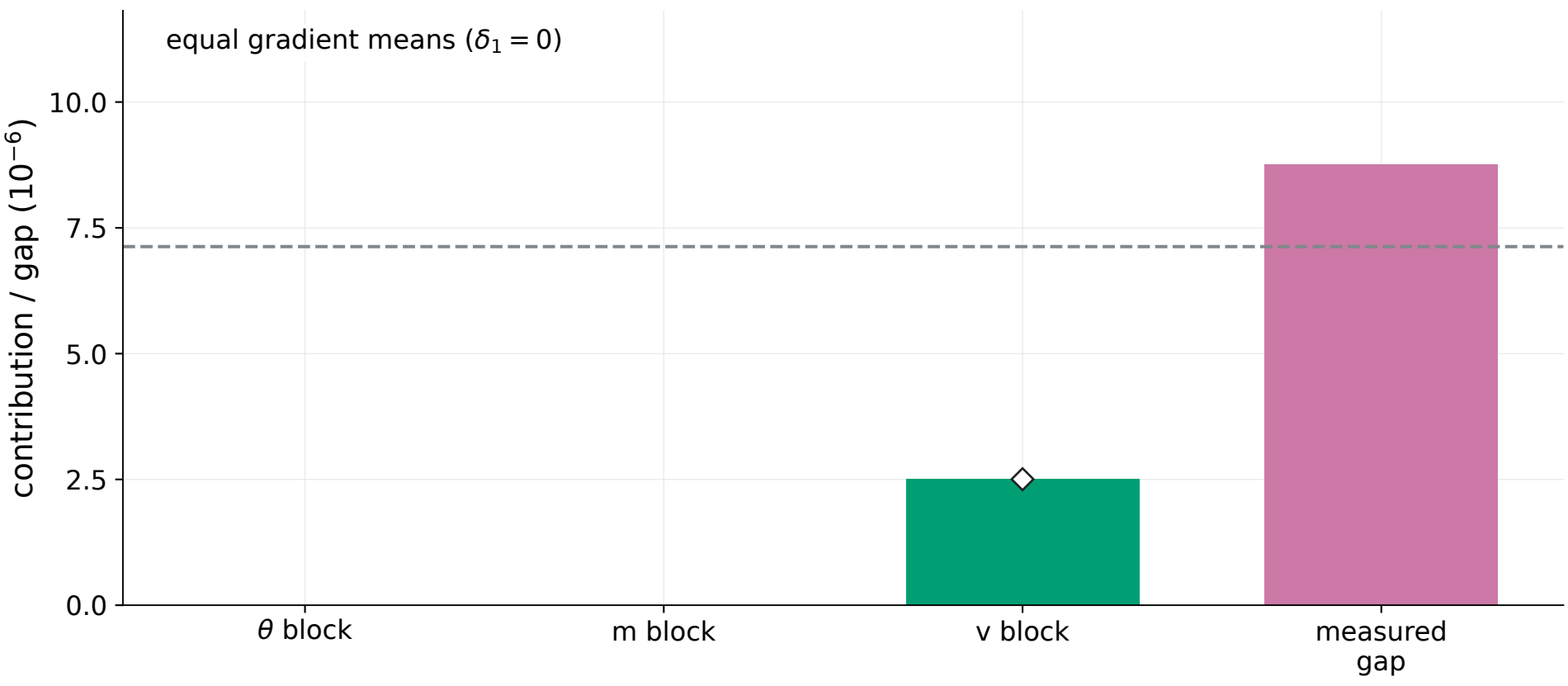
(a) Clipped SGD: the clipped mean decides



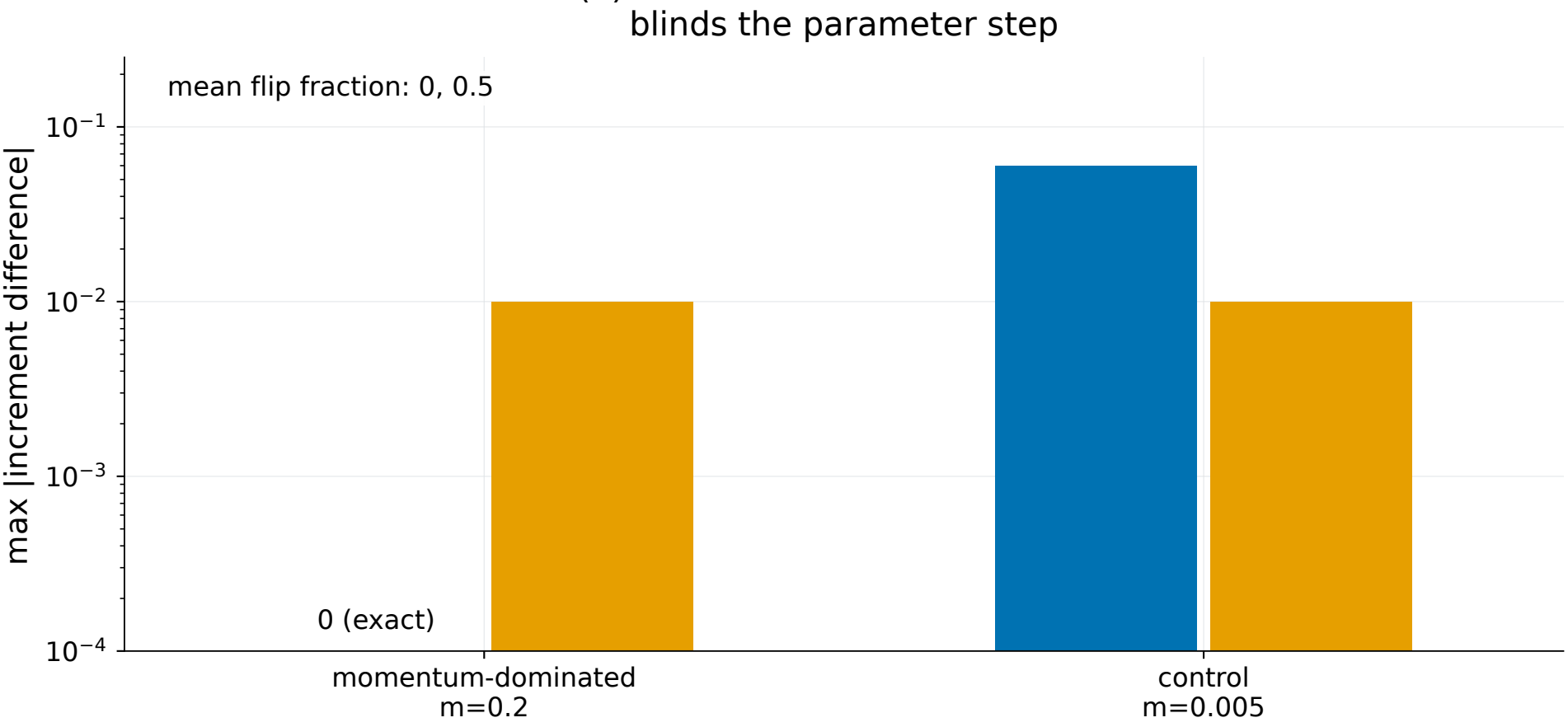
(b) Always-active clipping: only the direction matters



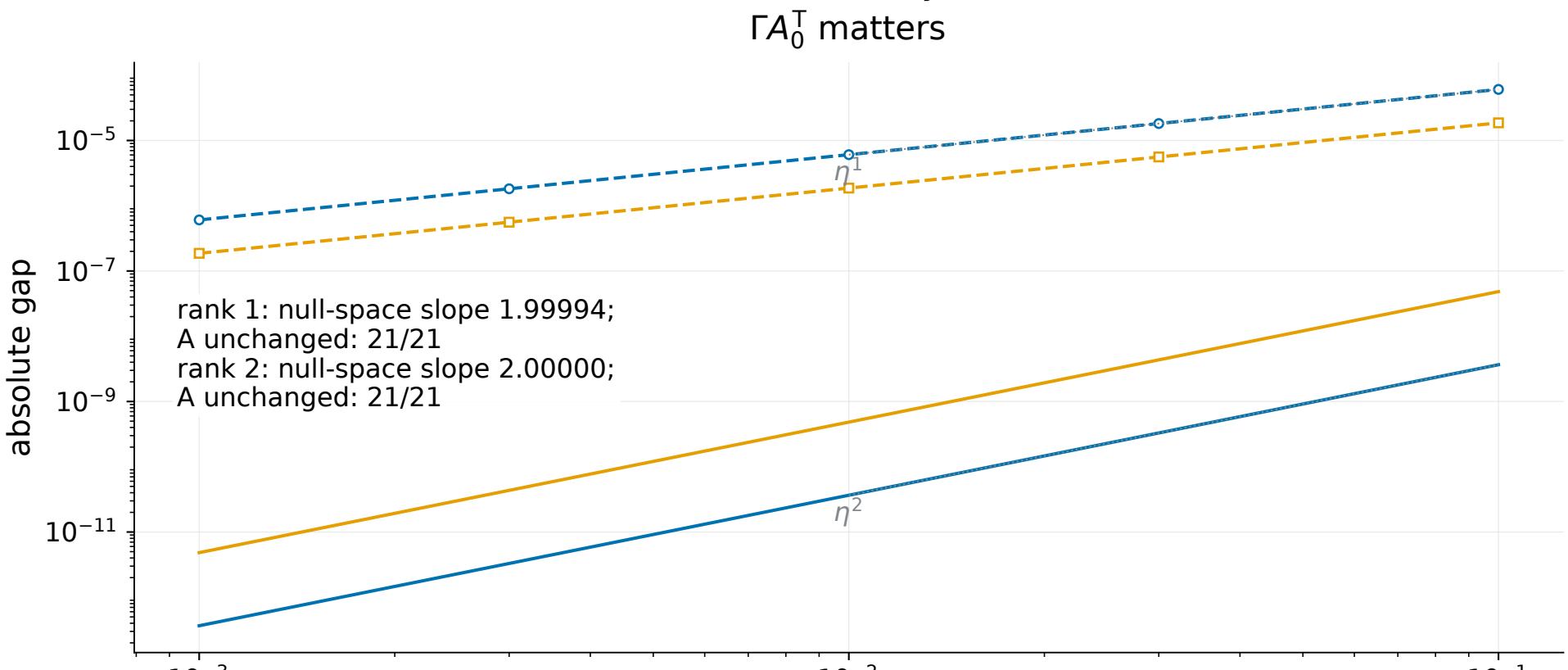
(c) Warm Adam: $E[G^2]$ enters through v



(d) Lion: momentum dominance blinds the parameter step



(e) LoRA at $B = 0$: only the sketch $\Gamma A_0^\top$ matters



(f) Learner → first-order statistic

SGD, momentum	$E\,G$
Adam (exact)	$E\,\Delta_{\text{Adam}}$
Adam (frozen denominator)	$E\,G$ and $E\,G^2$ (approximation)
cold Adam	sign law; $E\,G$, $E\,G^2$ in m, v
clipped SGD	clipped mean (direction if always active)
Lion	flip probability
LoRA, step 1	sketch $\Gamma A_0^\top$ (or its sign law)

Controlled small networks. The warm-Adam affine bound covers only the linear v contribution.